

Effect of Early vs Late Inguinal Hernia Repair on Serious Adverse Event Rates in Preterm Infants: A Randomized Clinical Trial

HIP Trial Investigators

IMPORTANCE Inguinal hernia repair in preterm infants is common and is associated with considerable morbidity. Whether the inguinal hernia should be repaired prior to or after discharge from the neonatal intensive care unit is controversial.

OBJECTIVE To evaluate the safety of early vs late surgical repair for preterm infants with an inguinal hernia.

DESIGN, SETTING, AND PARTICIPANTS A multicenter randomized clinical trial including preterm infants with inguinal hernia diagnosed during initial hospitalization was conducted between September 2013 and April 2021 at 39 US hospitals. Follow-up was completed on January 3, 2023.

INTERVENTIONS In the early repair strategy, infants underwent inguinal hernia repair before neonatal intensive care unit discharge. In the late repair strategy, hernia repair was planned after discharge from the neonatal intensive care unit and when the infants were older than 55 weeks' postmenstrual age.

MAIN OUTCOMES AND MEASURES The primary outcome was occurrence of any prespecified serious adverse event during the 10-month observation period (determined by a blinded adjudication committee). The secondary outcomes included the total number of days in the hospital during the 10-month observation period.

RESULTS Among the 338 randomized infants (172 in the early repair group and 166 in the late repair group), 320 underwent operative repair (86% were male; 2% were Asian, 30% were Black, 16% were Hispanic, 59% were White, and race and ethnicity were unknown in 9% and 4%, respectively; the mean gestational age at birth was 26.6 weeks [SD, 2.8 weeks]; the mean postnatal age at enrollment was 12 weeks [SD, 5 weeks]). Among 308 infants (91% with complete data (159 in the early repair group and 149 in the late repair group), 44 (28%) in the early repair group vs 27 (18%) in the late repair group had at least 1 serious adverse event (risk difference, -7.9% [95% credible interval, -16.9% to 0%]; 97% bayesian posterior probability of benefit with late repair). The median number of days in the hospital during the 10-month observation period was 19.0 days (IQR, 9.8 to 35.0 days) in the early repair group vs 16.0 days (IQR, 7.0 to 38.0 days) in the late repair group (82% posterior probability of benefit with late repair). In the prespecified subgroup analyses, the probability that late repair reduced the number of infants with at least 1 serious adverse event was higher in infants with a gestational age younger than 28 weeks and in those with bronchopulmonary dysplasia (99% probability of benefit in each subgroup).

CONCLUSIONS AND RELEVANCE Among preterm infants with inguinal hernia, the late repair strategy resulted in fewer infants having at least 1 serious adverse event. These findings support delaying inguinal hernia repair until after initial discharge from the neonatal intensive care unit.

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Group Information: The HIP Trial Investigators/Authors appear at the end of this article.

Corresponding Author: Martin L. Blakely, MD, MS, Department of Surgery, University of Texas Health Science Center, 7000 Fannin St, Houston, TX 77030 (martin.l.blakely@uth.tmc.edu).

Inguinal hernia is one of the most common congenital abnormalities requiring surgery in preterm infants. The incidence of inguinal hernia increases as gestational age decreases, reaching 40% in males born at 24 weeks' gestation.¹ Treatment is operative repair to prevent inguinal hernia incarceration (rate range, 5%-28%),^{2,3} but the safest timing for repair remains controversial.^{4,5} Preterm infants with inguinal hernia usually also have serious comorbidities (eg, acute respiratory distress syndrome, bronchopulmonary dysplasia, retinopathy of prematurity, and intraventricular hemorrhage),⁶ and undergoing the repair before discharge from the neonatal intensive care unit has raised concerns for surgical and anesthetic complications, possible prolonged mechanical ventilation, fragility of the tissues involved with the repair, and possible delayed discharge.⁴ Delaying the operation until after discharge from the neonatal intensive care unit possibly lowers those risks, but may increase hernia-related complications. The literature regarding optimal timing consists of retrospective analyses,⁷⁻¹¹ but no practice guideline recommendations.

The American Academy of Pediatrics highlighted the knowledge gap regarding optimal timing of inguinal hernia repair in preterm infants and emphasized the need for randomized clinical trials (RCTs).^{4,5} To address the controversy and knowledge gap, we conducted a multicenter RCT including preterm infants with inguinal hernia to compare the safety of early repair (prior to discharge from the neonatal intensive care unit) vs late repair (after discharge from the neonatal intensive care unit and when the infant was older than 55 weeks' postmenstrual age).

Methods

Trial Funding and Oversight

This RCT was conducted at 39 US medical centers and was funded by the Eunice Kennedy Shriver National Institute of Child Health and Human Development. The trial protocol was approved by the institutional review board at each study site and written informed consent was obtained from the parents or guardians before enrollment (the trial protocol appears in [Supplement 1](#)). The data and safety monitoring committee included national leaders in neonatology, pediatric surgery, and clinical trial statistics.

Patients

Infants were eligible for inclusion if they had a diagnosis of inguinal hernia, an estimated gestational age younger than 37 weeks, and were treated in a participating neonatal intensive care unit. Infants were excluded if (1) there were factors affecting the timing of hernia repair, (2) they were undergoing another operative procedure and the inguinal hernia repair was a secondary procedure, (3) there were known major congenital or chromosomal abnormalities that would likely influence the neurodevelopmental outcomes, or (4) the family was unable to return for follow-up and late inguinal hernia repair.

To assess diversity in the study population, parent- or guardian-reported race and ethnicity were obtained from the elec-

Key Points

Question Does the timing of inguinal hernia repair influence the likelihood of serious adverse events among preterm infants?

Findings In this randomized clinical trial including preterm infants in the neonatal intensive care unit with an inguinal hernia, 28% in the early hernia repair group vs 18% in the late hernia repair group had at least 1 serious adverse event (risk difference, -7.9%).

Meaning Delaying inguinal hernia repair in preterm infants until after neonatal intensive care unit discharge and when infants were older than 55 weeks' postmenstrual age appears to reduce the likelihood of serious adverse events.

tronic medical record by research coordinators and the data were used to assign the infants to race and ethnicity categories.

Study Design and Interventions

This study was a parallel group RCT comparing 2 strategies for the timing of inguinal hernia repair. Early repair was defined as operative inguinal hernia repair performed before discharge from the neonatal intensive care unit. The pediatric surgery team consulted with the neonatology and anesthesia teams and scheduled the operation when the infant was in optimal condition, prior to discharge from the neonatal intensive care unit.

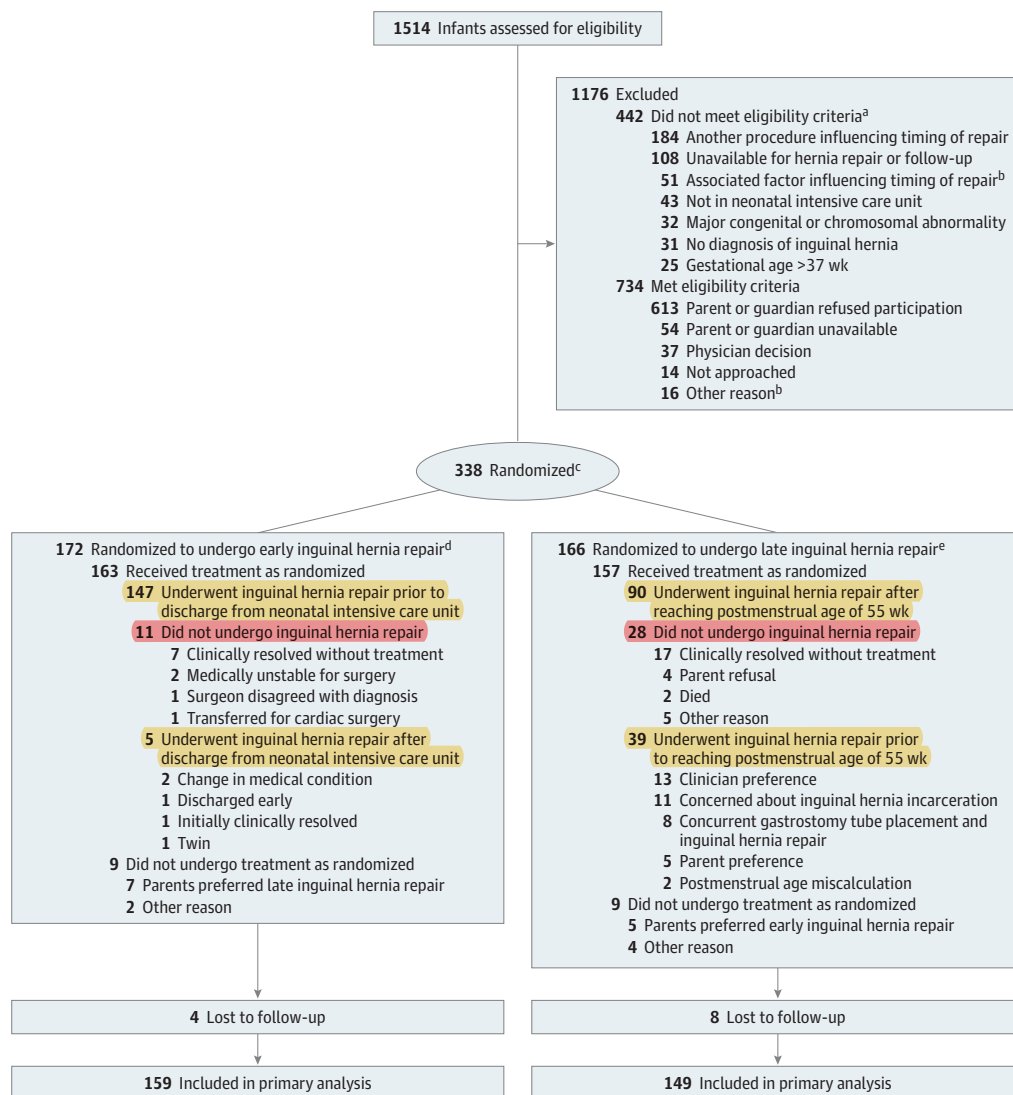
Late repair was defined as operative inguinal hernia repair performed after discharge from the neonatal intensive care unit and when the infant was older than 55 weeks' postmenstrual age (a threshold selected based on current pediatric anesthesia guidelines¹² that allows preterm infants to have elective outpatient operations when they are older than 55 weeks' postmenstrual age, given the decreased risk of anesthesia-related apnea or bradycardia). The pediatric surgery team counseled the family regarding signs of inguinal hernia incarceration, scheduled an in-person follow-up visit for 1 month after discharge from the neonatal intensive care unit, and made plans for hernia repair when the infant was older than 55 weeks' postmenstrual age.

Not all infants randomized to the early repair group were expected to undergo the operation before discharge from the neonatal intensive care unit because changes in their medical condition might have warranted a delay. Similarly, the timing of the operation in the infants randomized to the late repair group was expected to vary depending on the infant's condition and the concerns or availability of the parents or guardians and the surgeons. All participating study sites were large pediatric health care centers and all surgeons were pediatric general and thoracic surgeons.

Randomization Procedures

The infants were randomly assigned when approximately 2 weeks away from anticipated discharge from the neonatal intensive care unit in a 1:1 ratio to either early or late inguinal hernia repair ([Figure 1](#)). This enrollment time corresponds to when decisions are made regarding the timing of inguinal hernia repair in routine clinical practice. The random allocation sequence was created by the primary trial statistician

Figure 1. Flow Diagram of Infants Through the Trial of Early vs Late Inguinal Hernia Repair



^a There may have been more than 1 reason for an infant not meeting eligibility criteria.

^b Additional information appears in eTable 1 in Supplement 3.

^c Stratified by study site and gestational age categories (<28 weeks or ≥28 weeks).

^d Planned to be performed prior to discharge from the neonatal intensive care unit.

^e Planned to be performed after discharge from the neonatal intensive care unit and when the infant was older than 55 weeks' postmenstrual age.

and uploaded into the Research Electronic Data Capture (REDCap) database. Randomization was performed using the continuously available REDCap module¹³ in permuted blocks, with block sizes ranging from 2 to 4, and was stratified by center and gestational age (<28 weeks or ≥28 weeks).

Research coordinators were trained regarding REDCap randomization and data entry prior to study initiation. Screening data were entered for potentially eligible infants. When eligibility was confirmed and written informed consent was obtained, REDCap automatically assigned a unique study identification number and the infant was randomized to 1 of the intervention groups. The specific treatment group allocation was concealed until the infants were randomized

by the REDCap module. Data were collected for 10 months after randomization.

Outcomes

The primary outcome was occurrence of any prespecified serious adverse event during the 10-month observation period as determined by a blinded adjudication committee with expertise in pediatric surgery and anesthesia. The serious adverse events included pulmonary events (apnea requiring intervention, prolonged intubation, unplanned reintubation, stridor, pneumonia), cardiac events (bradycardia requiring intervention, cardiopulmonary resuscitation, cardiac arrest), surgical events (intraoperative injury, wound disruption, surgical site

infection), events related to the hernia (incarceration, recurrence, reoperation), and death.

Study personnel reviewed the electronic medical record daily for inpatients and weekly for outpatients to assess for possible adverse events. Families were contacted by study personnel for assessments 30 days after inguinal hernia repair and 10 months after enrollment. The adjudication committee reviewed the REDCap data for all infants who had an adverse event reported by site research coordinators at meetings facilitated by an overall trial coordinator. These reports included excerpts from the electronic medical record describing how the adverse events affected the infants and any subsequent interventions required.

The secondary outcomes were the number of days in the hospital during the 10-month observation period (included the initial neonatal intensive care unit stay after randomization, postoperative hospitalization, and any inpatient days due to hospital readmission) and the individual serious adverse events. The 2-year neurocognitive assessment is ongoing and investigators are using the Bayley Scales of Infant Development (third edition) to determine the cognitive composite score at 2 years' corrected age.

Statistical Analysis

The use of frequentist and bayesian analyses was prespecified in the trial protocol (Supplement 1). These 2 types of analyses were used to increase clarity of trial interpretation (as recommended by the American Statistical Association and others^{14,15}). The bayesian analyses were used primarily. The statistical analysis plan appears in Supplement 2.

The a priori primary hypothesis (informed by existing literature) was that the rate of serious adverse events would be 30% in the early inguinal hernia repair group and 20% in the late inguinal hernia repair group. A sample size of 586 infants with primary outcome data was required to have 80% power (2-sided α of .05) to detect a 10% absolute difference in the primary outcome. Assuming a lost to follow-up rate of 5% for the primary outcome, enrollment of 615 infants was planned. The intention-to-treat analyses were performed in the full dataset, and included all randomized infants except those withdrawn from the study (withdrawal was requested by the parents or guardians).

A logistic mixed-effects model was used to analyze the primary outcome, and a negative binomial mixed model was used for the secondary outcome of total days in the hospital during the observation period. All models included gestational age group as a covariate and study site as a random intercept. The frequentist and bayesian analyses used the same models; however, the frequentist analysis of the primary outcome used a generalized estimating equation logistic model (exchangeable correlation for study site) due to nonconvergence of the mixed-effects model.

All bayesian analyses used neutral priors. For the categorical and count outcomes, we assumed a prior centered at an odds ratio or relative risk (RR) of 1.0 with a 95% credible interval (CrI) of 0.33 to 3.0 for the intervention effect with vague priors for all other terms. Prespecified investigation of the treatment effect modifiers for the primary and secondary

outcomes were completed for the gestational age categories of younger than 28 weeks and 28 weeks or older, diagnosis of bronchopulmonary dysplasia, and open vs laparoscopic operative approach. These analyses used bayesian hierarchical models with main effects and interaction terms for the intervention groups and a modifier variable (1 variable at a time) using a prior centered at 1.0 (95% CrI, 0.33-3.0) for all 3 interaction terms.

For the primary outcome, the RRs and risk differences were estimated from the posterior distribution of the fitted logistic model. No missing outcomes were imputed. All statistical analyses were performed using R version 4.2.3 (R Foundation for Statistical Computing).

A formal interim analysis was completed when the data were available for 309 infants in February 2021. This interim analysis found a 97% probability of a decreased rate of serious adverse events in the late inguinal hernia repair group, which exceeded a prespecified efficacy stopping threshold of 95% and trial enrollment was stopped (as recommended by the data and safety monitoring committee).

Results

Patient Characteristics

Of the 1072 eligible infants, 338 (32%) were randomized between September 2013 and April 2021 (Figure 1). Among the 734 infants who met eligibility criteria but were not randomized, the most common reason for nonparticipation was parent or guardian refusal (84%; $n = 613$). Of these 613 parents or guardians, 280 preferred early repair, 196 preferred late repair, 71 wanted the clinician to decide, and 66 had other reasons (additional enrollment details appear in eTables 1-2 and details of the randomization by study site appear in eTable 3 in Supplement 3). Follow-up was completed on January 3, 2023.

The study population was predominantly male (86%), most had extremely low birth weights (median, 820 g [IQR, 640-1040 g]), and many had comorbidities (eg, 50% had bronchopulmonary dysplasia) (Table 1). Slightly more infants in the early repair group had apnea requiring intervention prior to enrollment (85%; $n = 137$) compared with the late repair group (79%; $n = 122$). Otherwise, there were no clinically important differences between the treatment groups, including the number of infants with a baseline gestational age younger than 28 weeks and those with bronchopulmonary dysplasia.

After randomization, 9 infants were withdrawn from each treatment group (they were withdrawn by the parents or guardians because of treatment preferences or for other reasons), leaving 163 in the early inguinal hernia repair group and 157 in the late inguinal hernia repair group. With 4 infants lost to follow-up in the early repair group and 8 infants lost to follow-up in the late repair group, primary outcome data were available for 308 infants (159 in the early repair group and 149 in the late repair group).

Surgery Characteristics

Among 163 infants in the early inguinal hernia repair group, 152 (93%) underwent hernia repair at a median postmenstrual age

Table 1. Baseline Characteristics of Infants by Early vs Late Inguinal Hernia Repair Group

	Infants, No. (%) ^a	
	Early inguinal hernia repair (n = 163) ^b	Late inguinal hernia repair (n = 157) ^c
Gestational age at birth, wk		
Median (IQR)	26 (25-28)	26 (24-28)
Mean (SD)	27 (3)	27 (3)
Gestational age		
<28 wk	103 (63)	104 (66)
≥28 wk	60 (37)	53 (34)
Age		
Median (IQR), d	80 (61-112)	82 (64-104)
Postnatal, mean (SD), wk	12 (5)	12 (5)
Postmenstrual, median (IQR), wk	39 (37-42)	39 (37-41)
Birth weight, median (IQR), g	835 (615-1072)	810 (660-1010)
Sex		
Male	141 (87)	133 (85)
Female	22 (13)	24 (15)
Race ^d		
Asian	5 (3)	3 (2)
Black	49 (28)	51 (31)
Native Hawaiian or Other Pacific Islander	1 (0.6)	0
White	99 (58)	101 (61)
Unknown	18 (10)	11 (7)
Ethnicity ^d		
Hispanic	28 (16)	26 (16)
Non-Hispanic	135 (78)	135 (81)
Unknown	9 (5)	5 (3)
Born at participating medical center ^e	98 (60)	92 (59)
Prior to enrollment		
Apnea requiring intervention ^f	137 (85)	122 (79)
Bronchopulmonary dysplasia ^g	82 (50)	78 (50)
Bradycardia requiring intervention ^h	72 (44)	70 (45)
Inguinal hernia incarceration	2 (1)	3 (2)
Intraventricular hemorrhage (grade 3 or 4) ⁱ	11 (7)	11 (7)
Diagnosis of inguinal hernia made by a pediatric surgeon rather than a neonatologist	115 (71)	112 (71)

^a Unless otherwise indicated. Excludes 9 infants (in each group) who were withdrawn after randomization.

^b Planned to be performed prior to discharge from the neonatal intensive care unit.

^c Planned to be performed after discharge from the neonatal intensive care unit and when the infant was older than 55 weeks' postmenstrual age.

^d Reported by the parent or guardian and entered in the electronic medical record. The denominators are 172 for the early repair group and 166 for the late repair group (these are the number of infants randomized to each group prior to withdrawals and those lost to follow-up).

^e Rather than having been born at another medical center and then transferred.

^f Included stimulation; initiation of nasal cannula oxygen, continuous positive airway pressure, or mechanical ventilation; and stimulant medication. Missing data for 1 infant in the early repair group and 2 infants in the late repair group.

^g Defined by physician documentation in the electronic medical record.

^h Included chest compressions and use of atropine. Missing data for 1 infant in each hernia repair group.

ⁱ Determined using cranial ultrasound (range, grade 1-4; higher grades are worse). Missing data for 1 infant in the early repair group. Grade 3 indicates intraventricular hemorrhage filling more than 50% of the ventricle and causing ventricular enlargement. Grade 4 indicates periventricular echo density.

of 41 weeks (IQR, 39-44 weeks) and had a median weight of 3.1 kg (IQR, 2.5-3.6 kg) at hernia repair. The reasons for 11 infants not undergoing early inguinal hernia repair included clinical resolution of the hernia (n = 7), medically unstable for surgery (n = 2), diagnostic uncertainty (n = 1), and infant transferred to another facility (n = 1). Of the 147 infants who underwent inguinal hernia repair prior to discharge from the neonatal intensive care unit, it was performed laparoscopically in 62 (42%) and under general anesthesia in 146 (99%). Five infants in the early repair group underwent the inguinal hernia repair after dis-

charge from the neonatal intensive care unit due to a change in their medical condition (n = 2) and other reasons (n = 3).

Among the 157 infants in the late inguinal hernia repair group, 129 (82%) underwent hernia repair at a median postmenstrual age of 57 weeks (IQR, 52-61 weeks) and had a median weight of 5.9 kg (IQR, 4.7-6.9 kg) at hernia repair. The reasons for 28 infants not undergoing inguinal hernia repair included clinical resolution of the hernia (n = 17), refusal by a parent or guardian (n = 4), unrelated death (n = 2), and other reasons (n = 5). Of the 129 infants who underwent inguinal

Table 2. Primary and Secondary Outcomes of Infants by Early vs Late Inguinal Hernia Repair Group^a

	Early inguinal hernia repair (n = 159) ^b	Late inguinal hernia repair (n = 149) ^c	Absolute risk difference (95% CrI), % ^d	Relative risk (95% CrI)	Posterior probability of benefit, % ^e
Primary outcome					
Had ≥1 serious adverse event, No. (%) ^f	44 (28)	27 (18)	-7.9 (-16.9 to 0)	0.68 (0.45 to 1.01)	97
Secondary outcome					
Hospital stay, median (IQR), d ^g	19.0 (9.8 to 35.0)	16.0 (7.0 to 38.0)	NA	0.91 (0.74 to 1.11)	82

Abbreviations: CrI, credible interval; NA, not applicable.

^a Excludes 9 infants (in each group) who were withdrawn after randomization and 12 who were lost to follow-up (4 in the early repair group and 8 in the late repair group). All analytic models included gestational age group and study site as covariates; there were no missing data for these variables.

^b Planned to be performed prior to discharge from the neonatal intensive care unit.

^c Planned to be performed after discharge from the neonatal intensive care unit and when the infant was older than 55 weeks' postmenstrual age.

^d Derived from bayesian logistic models including early vs late repair strategy as a covariate with study site as a random intercept.

^e Late repair relative to early repair (risk difference, <0; relative risk, <1; and fewer hospital days).

^f The specific events appear in Table 3.

^g From randomization until 10 months later (observation period). Data were missing for 3 infants in the early repair group.

hernia repair in the late repair group, it was performed laparoscopically in 47 (36%) and under general anesthesia in 127 (98%). There were 39 patients in the late repair group who underwent inguinal hernia repair when younger than 55 weeks' postmenstrual age; the reasons for early repair included clinician preference (n = 13), concerns for inguinal hernia incarceration (n = 11), placement of gastrostomy tube and inguinal hernia repair occurred concurrently (n = 8), parent preference (n = 5), and postmenstrual age miscalculation (n = 2).

The median duration of the inguinal hernia repair was 57 minutes (IQR, 41.8-90.2 minutes) in the early repair group and 70 minutes (IQR, 44.0-99.0 minutes) in the late repair group. More than 58% of infants who underwent inguinal hernia repair also underwent a concurrent operation (circumcision, placement of gastrostomy tube, umbilical hernia repair, or orchiopexy). More infants in the late repair group underwent orchiopexy compared with the early repair group (4.5% vs 1.2%, respectively).

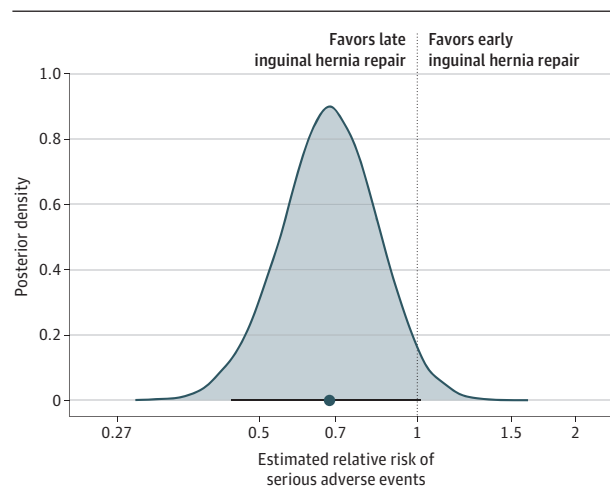
Primary Outcome

Among the 308 infants with complete data, 44 of 159 (28%) in the early repair group experienced at least 1 serious adverse event during the 10-month observation period vs 27 of 149 (18%) in the late repair group (absolute risk difference, -7.9% [95% CrI, -16.9% to 0%]; RR, 0.68 [95% CrI, 0.45 to 1.01]; 97% posterior probability of benefit with late repair for both the risk difference and RR; Table 2 and Figure 2). The frequentist analyses were consistent with these findings (eTable 2 in Supplement 3).

Secondary Outcomes

The median number of days in the hospital during the 10-month observation period was 19.0 days (IQR, 9.8-35.0 days) in the early repair group vs 16.0 days (IQR, 7.0-38.0 days) in the late repair group (RR, 0.91 [95% CrI, 0.74-1.11]; 82% posterior probability of benefit with late repair) (Table 2). From enrollment to initial discharge from the neonatal intensive care unit, the median number of days in the hospital was 18.0 days (IQR, 8.0-31.5 days) with early repair vs 13.0 days (IQR, 5.0-36.5 days) with late repair. The median time from inguinal hernia repair to discharge was 6.0 days (IQR, 3.0-11.5 days) for the

Figure 2. Posterior Probability of the Estimated Relative Risk of Serious Adverse Events



The blue dot and horizontal line indicate the median value and the percentile-based 95% credible interval (CrI). Early inguinal hernia repair was planned to be performed prior to discharge from the neonatal intensive care unit. Late inguinal hernia repair was planned to be performed after discharge from the neonatal intensive care unit and when the infant was older than 55 weeks' postmenstrual age. The estimated relative risk was 0.68 (95% CrI, 0.45-1.01) for the posterior distribution. The probability was 97% that the late repair strategy was associated with a reduced rate for any serious adverse event (to any extent) compared with the early repair strategy.

early repair group vs 0.5 (IQR, 0.5-1.0 days) for the late repair group. There were few hospital readmissions (35 [22%] in the early repair group vs 42 [28%] in the late repair group) and the median number of days in the hospital was 0 for both groups.

In the early repair group, the most common serious adverse events were apnea requiring increased respiratory intervention, prolonged intubation (>48 hours), and bradycardia requiring pharmacological intervention (Table 3). In the late repair group, the most common serious adverse events were apnea requiring increased respiratory intervention, inguinal hernia incarceration, bradycardia requiring pharmacological intervention, and cardiopulmonary resuscitation. There were no injuries to spermatic cord structures in any patients, but 2

Table 3. Serious Adverse Events Experienced by Infants in the Early vs Late Inguinal Hernia Repair Group

	Serious adverse event, No. (%) ^a	
	Early inguinal hernia repair (n = 159) ^b	Late inguinal hernia repair (n = 149) ^c
Apnea requiring intervention ^d	28 (17.6)	9 (6.0)
Prolonged intubation (>48 h)	6 (3.8)	0
Bradycardia requiring intervention ^e	5 (3.1)	4 (2.7)
Cardiopulmonary resuscitation	4 (2.5)	4 (2.7)
Stridor	3 (1.9)	1 (0.7)
Death	2 (1.3)	3 (2.0)
Pneumonia	2 (1.3)	0
Regional anesthesia toxicity	1 (0.6)	0
Unplanned reintubation	0	1 (0.7)
Cardiac arrest	0	3 (2.0)
Inguinal hernia		
Incarceration ^f	2 (1.3)	6 (4.0)
Recurrence	2 (1.3)	1 (0.7)
Reoperation	2 (1.3)	3 (2.0)
Intraoperative injury to adjacent structure	2 (1.3)	0
Wound disruption	0	1 (0.7)
Surgical site infection	1 (0.6)	2 (1.3)
Other event	1 (0.6) ^g	2 (1.3) ^h

^a An infant may have experienced more than 1 of the serious adverse events.

^b Planned to be performed prior to discharge from the neonatal intensive care unit.

^c Planned to be performed after discharge from the neonatal intensive care unit and when the infant was older than 55 weeks' postmenstrual age.

^d Included increased oxygen administration and increase in respiratory support (eg, use of bag and mask ventilation) or the initiation or increase in mechanical ventilation.

^e Treated with a pharmacological intervention.

^f Defined as inguinal hernia that is nonreducible (1) and necessitates an urgent inguinal hernia repair; (2) or requires narcotic pain medication or sedation to facilitate inguinal hernia reduction; (3) or requires the pediatric surgery team.

^g Had a pneumothorax identified after inguinal hernia repair requiring high-frequency oscillator ventilation for 3 days.

^h One infant had an infected scrotal hematoma requiring drainage and the other had bleeding from a concurrent circumcision requiring overnight hospital admission.

patients in the early repair group had intraoperative intestinal injuries (one had a full thickness injury and another had a serosal tear).

Two infants in the early repair group and 3 in the late repair group underwent a second repair operation. Inguinal hernia incarceration occurred in 2 infants in the early repair group and in 6 infants in the late repair group. Of the 8 infants with inguinal hernia incarceration, 1 had a bowel injury requiring resection (in the early repair group), but none required an emergency operation. The inguinal hernia clinically resolved in 7 infants (4%) in the early repair group vs 17 (11%) in the late repair group, precluding the need for surgery.

Heterogeneity of Treatment Effect

The RR for the primary outcome was 0.61 (95% CrI, 0.39-0.94) with late repair in infants younger than 28 weeks' gestation (99% bayesian posterior probability of benefit). The RR for the primary outcome was 0.92 (95% CrI, 0.47-1.75) in infants with a gestational age of 28 weeks or older (61% posterior probability of benefit). The bayesian probability of a gestational age × treatment group interaction was 91%.

Among those with bronchopulmonary dysplasia, the RR for the primary outcome with late repair was 0.50 (95% CrI, 0.27-0.87) and the posterior probability of benefit was 99%. Among those without bronchopulmonary dysplasia, the RR for the primary outcome with late repair was 0.85 (95% CrI, 0.51-1.37) and

the posterior probability of benefit was 75%. The bayesian probability of a bronchopulmonary dysplasia × treatment group interaction was 95%. Open vs laparoscopic repair did not modify the effect of assignment to early repair vs late repair (Figure 3).

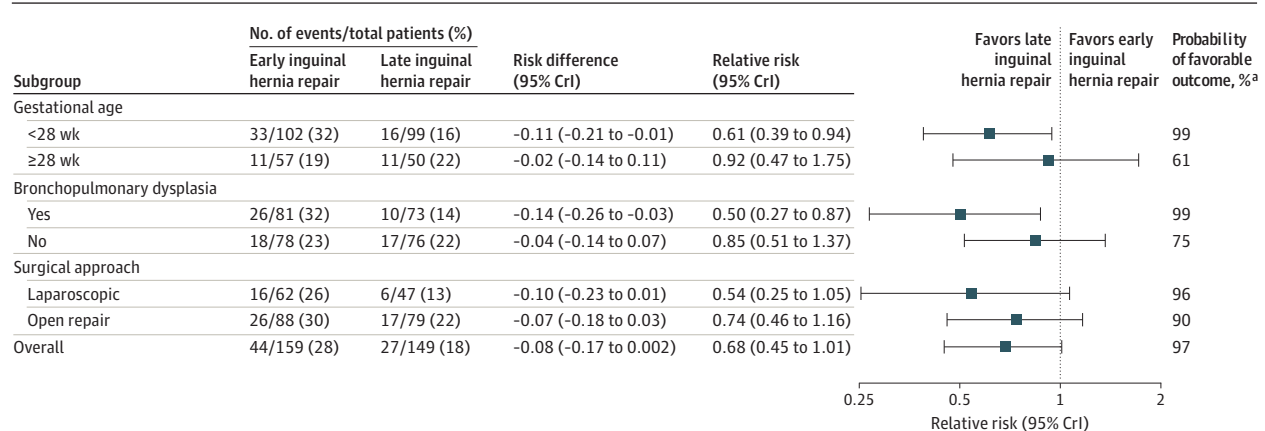
Discussion

In this RCT comparing early vs late inguinal hernia repair in preterm infants, those undergoing late repair had a lower probability of having at least 1 serious adverse event, spent fewer days in the hospital during the 10-month observation period, had a low incidence of inguinal hernia incarceration, and had a greater probability of hernia resolution without repair. The benefit of late repair was greatest in infants younger than 28 weeks' gestation and in those with bronchopulmonary dysplasia.

Serious adverse events among infants in the early repair group were usually related to anesthesia (apnea, prolonged intubation, bradycardia), which is consistent with prior reports. Infants treated with late repair had shorter stays in the neonatal intensive care unit after randomization and fewer postoperative days in the hospital.

The finding that hernias clinically resolved in 4% of infants in the early repair group and in 11% of infants in the late repair group supports later inguinal hernia repair and deserves further study. The incidence of closure of a patent

Figure 3. Subgroup Analyses of Serious Adverse Events Experienced by Infants in the Early vs Late Inguinal Hernia Repair Groups



Early inguinal hernia repair was planned to be performed prior to discharge from the neonatal intensive care unit. Late inguinal hernia repair was planned to be performed after discharge from the neonatal intensive care unit and when the infant was older than 55 weeks' postmenstrual age. The estimates were derived from bayesian logistic models including repair strategy (early vs late),

subgroup variable (1 variable at a time), and their interaction as covariates with study site as a random intercept. CrI indicates credible interval.

^a Late repair relative to early repair (risk difference, <0; relative risk, <1).

processus vaginalis (the etiology of infant inguinal hernia) has been estimated to be 60% in children up to 2 years of age,¹⁶ and other investigators have reported greater than 30% clinical resolution.^{17,18} Hernia resolution was determined with an examination by a surgeon because diagnostic imaging is not commonly used by pediatric surgeons to determine the need for an operation or even presence of inguinal hernia.

There are several important considerations for clinicians contemplating how these new data should influence their practice. Preterm infants with an inguinal hernia diagnosed during their initial neonatal intensive care unit hospitalization have typically been hospitalized for several months before the discussion on timing of inguinal hernia repair occurs, and parents have often developed preferences for early or late hernia repair.

The decision to treat the inguinal hernia with an early or late repair strategy likely does not influence the overall duration of the neonatal intensive care unit stay, but may hasten the discharge by several days if later repair is chosen, which is likely important to parents and neonatologists. The likelihood of an infant experiencing at least 1 serious adverse event is an important consideration and heretofore was not known. We believe the trial results are generalizable to most preterm infants with an inguinal hernia diagnosed during the initial neonatal intensive care unit stay, given the inclusive eligibility criteria and the large number of medical centers and diversity within the centers.

Limitations

This study has limitations. First, the modest sample size (338 randomized infants) limits the precision of the findings. The 32% randomization rate of eligible infants was lower than expected, but similar to other recently completed neonatal surgery RCTs.^{19,20} The trial was stopped early due to meeting a prespecified bayesian stopping rule for effectiveness, which limits the trial's size but is ethically justified.²¹

Second, infant withdrawal from the trial due to parent or guardian request, and the lack of data on these infants, further reduced the analysis sample, but is consistent with the ethical guidelines from the Office for Human Research Protections.²²

Third, the COVID-19 pandemic certainly increased obstacles for infant follow-up²³ after discharge from the neonatal intensive care unit and impaired research personnel attempting to acquire data in this trial.

Conclusions

Among preterm infants with inguinal hernia, the late repair strategy resulted in fewer infants having at least 1 serious adverse event. These findings support delaying inguinal hernia repair until after initial discharge from the neonatal intensive care unit.

ARTICLE INFORMATION

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HIP Trial Investigators/Authors: Martin L. Blakely, MD, MS; Andrea Krzyzaniak, MA; Melvin S. Dassinger, MD; Claudia Pedroza, PhD; Jorn-Hendrik Weitkamp, MD, PhD; Ankush Gosain, MD, PhD; Michael Cotten, MD; Susan R. Hintz, MD, MS; Henry Rice, MD; Sherry E. Courtney, MD; Kevin P. Lally, MD, MS; Namasivayam Ambalavanan, MD; Catherine M. Bendel, MD; Kim Chi T. Bui, MD; Casey

Calkins, MD; Nicole M. Chandler, MD; Roshni Dasgupta, MD, MPH; Jonathan M. Davis, MD; Katherine Deans, MD, MS; Daniel A. DeUgarte, MD; Jeffrey Gander, MD; Carl-Christian A. Jackson, MD; Martin Keszler, MD; Karen Kling, MD; Stephen J. Fenton, MD; Kimberley A. Fisher, PhD; Tyler Hartman, MD; Eunice Y. Huang, MD, MS; Saleem Islam, MBBS, MPH; Frances Koch, MD; Shabnam Lainwala, MD, PhD; Aaron Leshner, MD; Monica Lopez, MD, MS; Meghna Misra, MD; Jamie Overbey,

DO; Brenda Poindexter, MD, MS; Robert Russell, MD, MPH; Steven Stylianos, MD; Douglas Y. Tamura, MD; Bradley A. Yoder, MD; Donald Lucas, MD; Donald Shaul, MD; P. Ben Ham III, MD; Colleen Fitzpatrick, MD; Kara Calkins, MD; Aaron Garrison, MD; Diomed de la Cruz, MD; Shahab Abdessalam, MD; Charlotte Kvasnovsky, MD; Bradley J. Segura, MD, PhD; Joel Shilyansky, MD; Lynne M. Smith, MD; Jon E. Tyson, MD, MPH.

Affiliations of HIP Trial Investigators/Authors:

Department of Surgery, Institute for Clinical Research and Learning Healthcare and Institute for Implementation Science, University of Texas Health Science Center, Houston (Blakely); Scripps Mercy Hospital, San Diego, California (Krzyzaniak); Division of Pediatric Surgery, University of Arkansas for Medical Sciences, Little Rock (Dassinger); Department of Pediatrics, Institute for Clinical Research and Learning Healthcare, University of Texas Health Science Center, Houston (Pedroza, Tyson); Division of Neonatology, Vanderbilt University Medical Center, Nashville, Tennessee (Weitkamp); Division of Pediatric Surgery, University of Colorado, Aurora (Gosain); Division of Neonatology, Duke University, Durham, North Carolina (Cotten, Fisher); Division of Neonatology, Stanford University, Palo Alto, California (Hintz); Division of Pediatric Surgery, Duke University, Durham, North Carolina (Rice); Division of Neonatology, University of Arkansas for Medical Sciences, Little Rock (Courtney); Department of Pediatric Surgery, University of Texas Health Science Center, Houston (Lally); Division of Neonatology, University of Alabama at Birmingham (Ambalavanan); Division of Neonatology, University of Minnesota, Minneapolis (Bendel); Division of Neonatology, Kaiser Permanente, Los Angeles, California (Bui); Division of Pediatric Surgery, Medical College of Wisconsin, Milwaukee (C. Calkins); Division of Pediatric Surgery, Johns Hopkins All Children's Hospital, St Petersburg, Florida (Chandler); Division of Pediatric General and Thoracic Surgery, Cincinnati Children's Hospital Medical Center, Cincinnati, Ohio (Dasgupta, Garrison); Division of Neonatology, Tufts Medical Center, Boston, Massachusetts (Davis); Department of Pediatric Surgery, Nemours Children's Hospital, Wilmington, Delaware (Deans); Division of Pediatric Surgery, David Geffen School of Medicine, University of California, Los Angeles (DeUgarte); Division of Pediatric Surgery, University of Virginia, Charlottesville (Gander); Division of Pediatric Surgery, Alpert Medical School, Brown University, Providence, Rhode Island (Jackson); Division of Neonatology, Alpert Medical School, Brown University, Providence, Rhode Island (Keszler); Rady Children's Hospital and Division of Pediatric Surgery, University of California, San Diego (Kling); Division of Pediatric Surgery, University of Utah, Salt Lake City (Fenton); Division of Neonatology, Geisel School of Medicine, Dartmouth College, Hanover, New Hampshire (Hartman); Department of Pediatric Surgery, Vanderbilt University Medical Center, Nashville, Tennessee (Huang, Lopez); Division of Pediatric Surgery, University of Florida, Gainesville (Islam); Department of Surgery, Aga Khan University, Sindh, Pakistan (Islam); Division of Neonatology, Medical University of South Carolina, Charleston (Koch); Division of Neonatology, Connecticut Children's Medical Center, Hartford (Lainwala); Division of Pediatric Surgery, Medical University of South Carolina, Charleston (Leshner); Pediatric Surgery, Elliot Hospital, Manchester, New Hampshire (Misra); Division of Neonatology, Naval Medical Center, San Diego, California (Overbey); Division of Neonatology, School of Medicine, Emory University, Atlanta, Georgia (Poindexter); Division of Pediatric Surgery, University of Alabama at Birmingham (Russell); Division of Pediatric Surgery, Columbia University Medical Center, Morgan Stanley Children's Hospital, New York, New York (Stylianos); Division of

Pediatric Surgery, Valley Children's Hospital, Madera, California (Tamura); Division of Neonatology, University of Utah, Salt Lake City (Yoder); F. Edward Hebert School of Medicine, Uniformed Services University of the Health Sciences, Bethesda, Maryland (Lucas); Division of Pediatric Surgery, Naval Medical Center, San Diego, California (Lucas); Division of Pediatric Surgery, Kaiser Permanente, Los Angeles, California (Shaul); Division of Pediatric Surgery, University at Buffalo, Buffalo, New York (Ham III); Division of Pediatric Surgery, Cohen Children's Medical Center, Northwell Health, New Hyde Park, New York (Fitzpatrick); Division of Neonatology, David Geffen School of Medicine, University of California, Los Angeles (K. Calkins); Division of Neonatology, School of Medicine, University of Florida, Gainesville (de la Cruz); Division of Neonatology, University of Nebraska Medical Center, Omaha (Abdessalam); Division of Pediatric Surgery, University of Chicago, Chicago, Illinois (Kvasnovsky); Division of Pediatric Surgery, M Health Fairview University of Minnesota Masonic Children's Hospital, Minneapolis (Segura); Department of Pediatric Surgery, University of Iowa Stead Family Children's Hospital, Iowa City (Shilyansky); Harbor-UCLA Medical Center, Torrance, California (Smith).

Author Contributions: Dr Blakely had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Blakely, Krzyzaniak, Dassinger, Pedroza, Hintz, Courtney, Lally, Segura, Tyson.

Acquisition, analysis, or interpretation of data:

Blakely, Krzyzaniak, Dassinger, Pedroza, Weitkamp, Gosain, Cotten, Rice, Lally, Ambalavanan, Bendel, Bui, C. Calkins, Chandler, Dasgupta, Davis, Deans, DeUgarte, Gander, Jackson, Keszler, Kling, Fenton, Fisher, Hartman, Huang, Islam, Koch, Lainwala, Leshner, Lopez, Misra, Overbey, Poindexter, Russell, Stylianos, Tamura, Yoder, Lucas, Shaul, Ham, Fitzpatrick, K. Calkins, Garrison, de la Cruz, Abdessalam, Kvasnovsky, Segura, Shilyansky, Smith, Tyson.

Drafting of the manuscript: Blakely, Krzyzaniak, Pedroza, Gosain, Rice, Courtney, Lally, Dasgupta, Davis, Islam, Lopez, de la Cruz, Abdessalam, Kvasnovsky, Segura, Tyson.

Critical review of the manuscript for important intellectual content: Blakely, Krzyzaniak, Dassinger, Pedroza, Weitkamp, Gosain, Cotten, Hintz, Rice, Lally, Ambalavanan, Bendel, Bui, C. Calkins, Chandler, Dasgupta, Davis, Deans, DeUgarte, Gander, Jackson, Keszler, Kling, Fenton, Fisher, Hartman, Huang, Islam, Koch, Lainwala, Leshner, Lopez, Misra, Overbey, Poindexter, Russell, Stylianos, Tamura, Yoder, Lucas, Shaul, Ham, Fitzpatrick, K. Calkins, Garrison, de la Cruz, Abdessalam, Kvasnovsky, Segura, Shilyansky, Smith, Tyson.

Statistical analysis: Blakely, Krzyzaniak, Pedroza, Kvasnovsky, Segura, Tyson.

Obtained funding: Blakely, Ambalavanan, Segura, Tyson.

Administrative, technical, or material support: Krzyzaniak, Dassinger, Gosain, Rice, Courtney, Lally, Bui, C. Calkins, Chandler, Davis, Deans, DeUgarte, Jackson, Kling, Fisher, Hartman, Huang, Islam, Yoder, Lucas, Shaul, Ham, K. Calkins, Garrison, Abdessalam, Segura.

Supervision: Blakely, Krzyzaniak, Dassinger, Cotten, Lally, Dasgupta, Davis, Deans, DeUgarte, Kling,

Islam, Lainwala, Leshner, Overbey, Poindexter, Russell, Yoder, Ham, K. Calkins, Kvasnovsky, Segura, Smith, Tyson.

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REFERENCES

1. Ramachandran V, Edwards CF, Bichianu DC. Inguinal hernia in premature infants. *Neoreviews*. 2020;21(6):e392-e403. doi:10.1542/neo.21-6-e392
2. Wiener ES, Touloukian RJ, Rodgers BM, et al. Hernia survey of the section on surgery of the American Academy of Pediatrics. *J Pediatr Surg*. 1996;31(8):1166-1169. doi:10.1016/S0022-3468(96)90110-4
3. Antonoff MB, Kreykes NS, Saltzman DA, Acton RD. American Academy of Pediatrics section on surgery hernia survey revisited. *J Pediatr Surg*. 2005;40(6):1009-1014. doi:10.1016/j.jpedsurg.2005.03.018
4. Wang KS; Committee on Fetus and Newborn, American Academy of Pediatrics; Section on Surgery, American Academy of Pediatrics. Assessment and management of inguinal hernia in infants. *Pediatrics*. 2012;130(4):768-773. doi:10.1542/peds.2012-2008
5. Khan FA, Jancelewicz T, Kieran K, Islam S; Committee on Fetus and Newborn; Section on Surgery; Section on Urology. Assessment and management of inguinal hernias in children. *Pediatrics*. 2023;152(1):1-9. doi:10.1542/peds.2023-062510
6. Mowitz ME, Gao W, Sipsma H, et al. Burden of comorbidities and healthcare resource utilization among Medicaid-enrolled extremely premature infants. *J Health Econ Outcomes Res*. 2022;9(2):147-155. doi:10.36469/jheor.2022.38847

7. Lautz TB, Raval MV, Reynolds M. Does timing matter? a national perspective on the risk of incarceration in premature neonates with inguinal hernia. *J Pediatr*. 2011;158(4):573-577. doi:10.1016/j.jpeds.2010.09.047
8. Ferrantella A, Sola JE, Parreco J, et al. Complications while awaiting elective inguinal hernia repair in infants: not as common as you thought. *Surgery*. 2021;169(6):1480-1485. doi:10.1016/j.surg.2020.12.016
9. Gulack BC, Greenberg R, Clark RH, et al. A multi-institution analysis of predictors of timing of inguinal hernia repair among premature infants. *J Pediatr Surg*. 2018;53(4):784-788. doi:10.1016/j.jpedsurg.2017.09.009
10. Sacks MA, Neal D, Pairawan S, et al. Optimal timing of inguinal hernia repair in premature infants: a NSQIP-P study. *J Surg Res*. 2023;283:690-698. doi:10.1016/j.jss.2022.11.011
11. Sulkowski JP, Cooper JN, Duggan EM, et al. Does timing of neonatal inguinal hernia repair affect outcomes? *J Pediatr Surg*. 2015;50(1):171-176. doi:10.1016/j.jpedsurg.2014.10.035
12. Coté CJ, Zaslavsky A, Downes JJ, et al. Postoperative apnea in former preterm infants after inguinal herniorrhaphy: a combined analysis. *Anesthesiology*. 1995;82(4):809-822. doi:10.1097/0000542-199504000-00002
13. Harris PA, Taylor R, Thielke R, Payne J, Gonzalez N, Conde JG. Research Electronic Data Capture (REDCap)—a metadata-driven methodology and workflow process for providing translational research informatics support. *J Biomed Inform*. 2009;42(2):377-381. doi:10.1016/j.jbi.2008.08.010
14. Wasserstein RL, Lazar NA. *The ASA Statement on P Values: Context, Process, and Purpose*. Taylor & Francis; 2016:129-133.
15. Wasserstein RL, Schirm AL, Lazar NA. *Moving to a World Beyond "P<0.05"*. Taylor & Francis; 2019:1-19.
16. Rowe MI, Copelson LW, Clatworthy HW. The patent processus vaginalis and the inguinal hernia. *J Pediatr Surg*. 1969;4(1):102-107. doi:10.1016/0022-3468(69)90189-4
17. Fleming MA II, Grabski DF, Abebrese EL, et al. Clinical regression of inguinal hernias in premature infants without surgical repair. *Pediatr Surg Int*. 2021;37(9):1295-1301. doi:10.1007/s00383-021-04938-7
18. Oudesluys-Murphy AM, Teng HT, Boxma H. Spontaneous regression of clinical inguinal hernias in preterm female infants. *J Pediatr Surg*. 2000;35(8):1220-1221. doi:10.1053/jpsu.2000.8757
19. Blakely ML, Tyson JE, Lally KP, et al; Eunice Kennedy Shriver National Institute of Child Health and Human Development Neonatal Research Network. Initial laparotomy versus peritoneal drainage in extremely low birthweight infants with surgical necrotizing enterocolitis or isolated intestinal perforation: a multicenter randomized clinical trial. *Ann Surg*. 2021;274(4):e370-e380. doi:10.1097/SLA.0000000000005099
20. Blakely ML, Rysavy MA, Lally KP, Eggleston B, Pedroza C, Tyson JE. Special considerations in randomized trials investigating neonatal surgical treatments. *Semin Perinatol*. 2022;46(7):1516-40. doi:10.1016/j.semperi.2022.151640
21. Malmqvist E, Juth N, Lynöe N, Helgesson G. Early stopping of clinical trials: charting the ethical terrain. *Kennedy Inst Ethics J*. 2011;21(1):51-78. doi:10.1353/ken.2011.0002
22. Office for Human Research Protections; Department of Health and Human Services. Guidance on withdrawal of subjects from research: data retention and other related issues. Accessed February 29, 2024. <https://www.hhs.gov/ohrp/sites/default/files/ohrp/policy/subjectwithdrawal.pdf>
23. Tran NN, Tran M, Lopez J, Woon J, Nguyen J, Brecht ML. Impact of COVID-19 on pediatric clinical research. *J Pediatr Nurs*. 2021;61:240-244. doi:10.1016/j.pedn.2021.06.005